Qian 2020 Breast — analysis summary

RUN 1 OF 3

GO / NO-GO GATES

Gate 1 — mixing & continuity → NO-GO

Malignant cells form a per-patient archipelago. A single connected kNN graph appears only at the most generous integration (θ=10), where iLISI is just 1.1 of 7 samples — the scalar a naïve test reads as 'integrated' yet the embedding is organised BY PATIENT: 6/7 patients self-cluster far above chance (self-neighbour enrichment 6.8–23.5×, all past the 2.5× line). θ=5 splits the manifold into two disconnected components, one PURE sample 54 (the dominant 65% tumour). Integration does not fix it — enrichment GROWS with θ and is PC-count-invariant (22× at n_pcs 20/30/40), so Harmony over-corrects. Every patient carries the SAME embryonic modules — inter-patient CNV/clonal heterogeneity, not a shared cell-state continuum — do not force a trajectory.

Gate 2 — module geography → NA

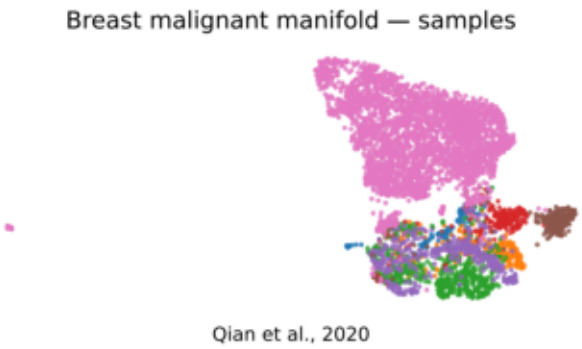
Not reached. Gate 2 tests a Gast/pSom hub and Crest geography on a fitted manifold; with Gate 1 NO-GO no manifold/trajectory is fit, so this gate does not apply.

Gate 3 — hub cross-repression → NA

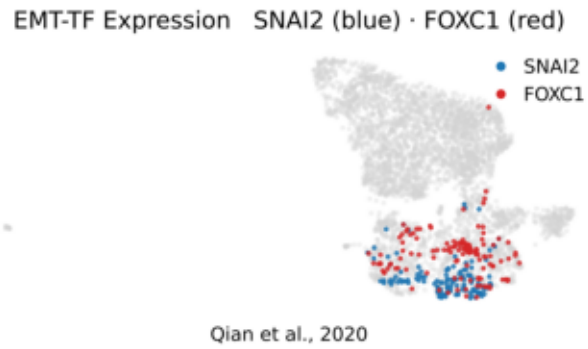
Not reached (branching-only, downstream of Gate 2). No hub, no bifurcation, no hub r.

STEP FIGURES

Manifold — samples



EMT-TF expression



Dataset.
Breast cancer (Qian et al., 2020), 10x (UMI counts) — 4,818 malignant cells across 7 samples = 7 patients. Gate 1 NO-GO: per-patient archipelago, no integrated continuum.

Verdict.
The embryonic-program pattern does NOT reproduce in Qian 2020 breast cancer: the malignant compartment is a per-patient archipelago — no sample-integrated, continuous cell-state manifold. One patient (sample 54, 65% of cells) even forms a fully disconnected component at moderate integration. Gate 1 NO-GO — analysis stops before the trajectory (Gates 2 & 3 N/A). EMT-TFs SNAI2 and FOXC1 are the broadest detectable factors, reported over the un-integrated manifold.

Parameter choices

Sample threshold	≥100 floor; argmax gap 3111→518 (6.01×) = dominant tumour, not a tail; keeps 7/7	**Gene / PCA**	1,225 overdispersed genes; n_pcs=20 (PC1 21% var, inter-patient/CNV axis)
Integration	Harmony on sample, swept θ 2/5/10/20/40; θ=10 first single component (diagnostic)	**Root / geometry**	none — no trajectory (Gate 1 NO-GO); per-patient archipelago
EMT-TFs shown	SNAI2 (11.3%) · FOXC1 (10.4%), overlaid without pseudotime		